MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, PIN CONTACTS, DECADE INCREMENTS 90, 100 AND 120 CONTACT POSITIONS FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

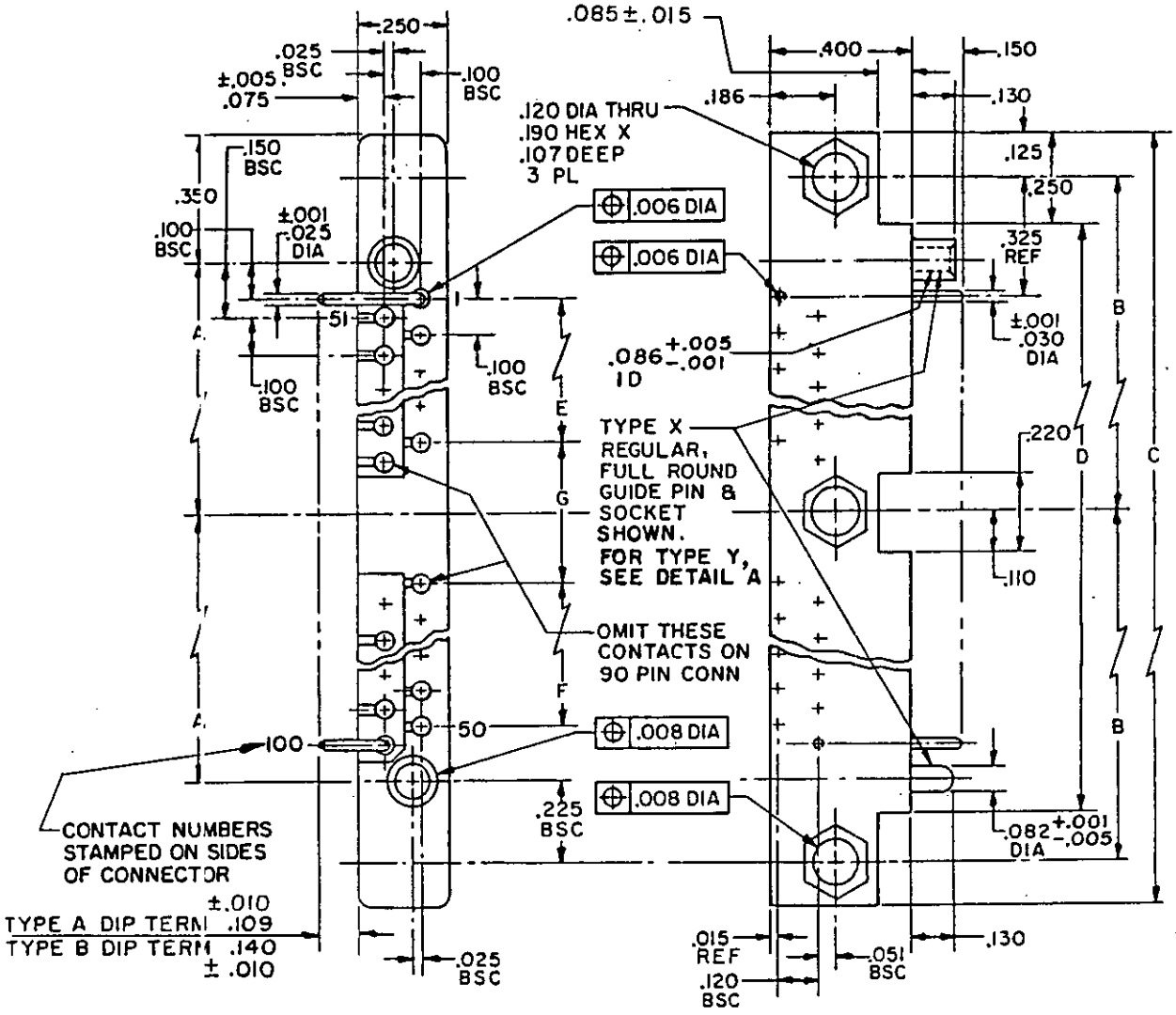


FIGURE 1. Connector, plug .100 (2.54 mm) spacing.

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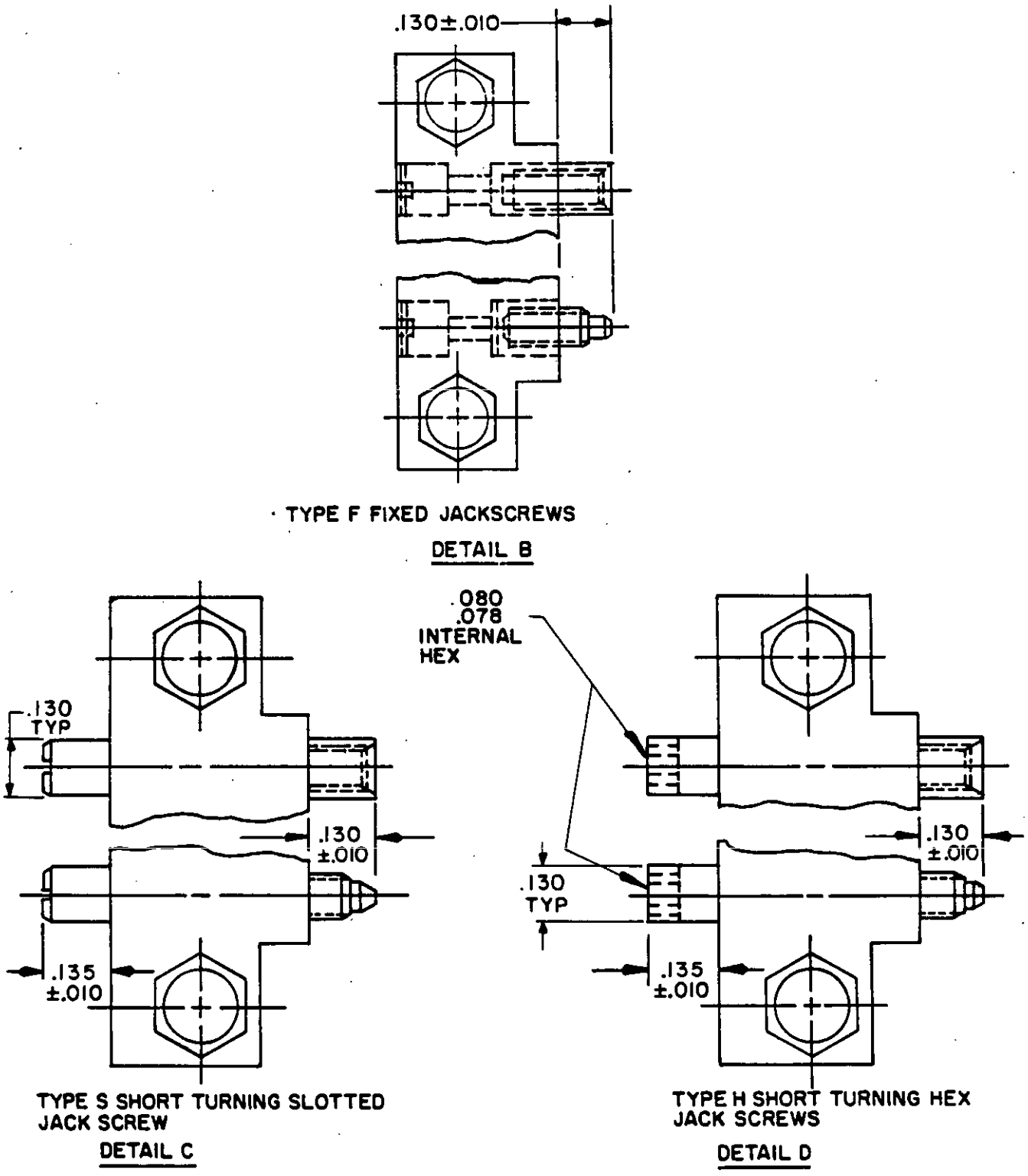
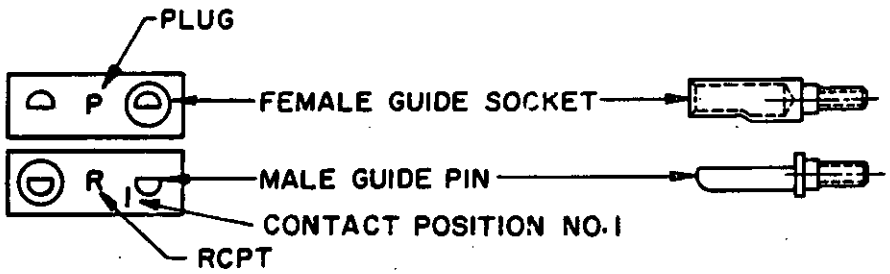


FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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CONNECTOR POLARIZATION WITH D SHAPED GUIDE PINS AND SOCKETS. (TYPE Y)

DETAIL A

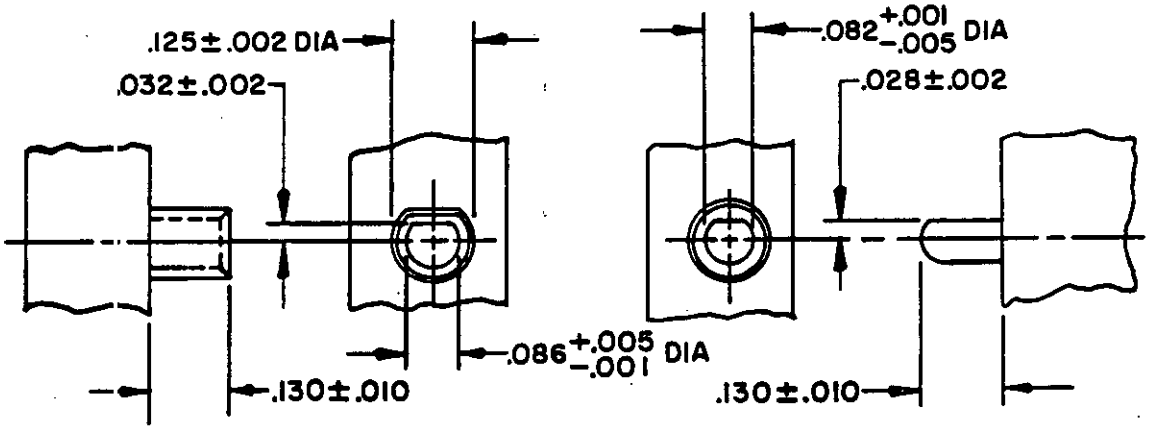


FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.001	.03	.032	.81	.110	2.79	.214	5.44
.002	.05	.051	1.30	.120	3.05	.220	5.59
.005	.13	.075	1.91	.125	3.18	.225	5.72
.006	.15	.078	1.98	.130	3.30	.235	5.97
.008	.20	.082	2.08	.135	3.43	.250	6.35
.010	.25	.086	2.18	.140	3.56	.350	8.89
.020	.51	.090	2.29	.150	3.81	.395	10.03
.025	.64	.100	2.54	.172	4.37	.400	10.16
.028	.71	.107	2.72	.190	4.83	.425	10.80
.030	.76	.109	2.77	.200	5.08		

Number of contacts	Dimensions						
	BSC A	BSC B	±.010 C	±.010 D	Ref E	Ref F	Ref G
90	2.475 (62.87)	2.700 (68.58)	5.650 (143.51)	5.150 (130.81)	2.200 (55.88)	2.100 (53.34)	.400 (10.16)
100	2.725 (69.22)	2.950 (74.93)	6.150 (156.21)	5.650 (143.51)	2.400 (60.95)	2.400 (60.95)	.400 (10.16)
120	3.225 (81.92)	3.450 (87.63)	7.150 (181.61)	6.650 (168.91)	2.900 (73.66)	2.900 (73.66)	.400 (10.16)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/60.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number 1 contact shall be marked.
6. Termination layout on .020 (0.51 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Optional undercut .020 maximum, on PCB side for cleaning purposes.

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion resisting stainless steel per ASTM-A581 or ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max. other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper per QQ-C-550, nickel plated per QQ-N-290, class 2, 50 to 150 microinches.

Plating: The contact plating shall be per MIL-C-55302 or gold per MIL-G-45204, type II, grade C, class 1, over nickel plating per QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold per MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating per QQ-N-290, class 02, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:
Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum per contact.

Polarization: Type X, regular, guide pin and socket, full round (see figure 1). Type Y, adjustable "D" shaped guide pin and/or socket (see MIL-C-55302/57, detail B). All connectors are supplied in the -1 polarized position. For other than OEM use, connectors may be re-polarized by using M55302/57-01 spanner wrench and M55302/57-02 guide pin and guide socket set. When guides are removed and reinstalled in another position apply Loctite 83-21 or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three (3) spaces after the part number on the purchase order. Example M55302/59-A10Y [-32]. Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench M55302/57-01. Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three (3) spaces after the part number on the purchase order. Guide sets, as installed, shall withstand 7 inch ounces backoff torque applied through the guide set mounting nut.

M55302/57-02 Kit: Consists of one type Y "D" shaped guide socket and one type Y "D" shaped guide pin.

Example of part number:

M55302/59 — Basic number of specification sheet

A — Type of terminal (see figure 1)
- A = dip terminal .109 long
- B = dip terminal .140 long
- C = dip terminal .172 long

100 — Number of contacts (see figure 1)

X — Type of mounting hardware (see details A,B,C, and D of figure 1)
- Type X = full round
- Type Y = "D" shaped
- Type F = fixed jackscrew
- Type S = .135 slotted turning jackscrew
- Type H = .135 hex turning jackscrew

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3592-5)

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